

Amino acid sensing in dietary-restriction-mediated longevity: roles of signal-transducing kinases GCN2 and TOR

DR (dietary restriction), or reduced food intake without malnutrition, is associated with extended longevity, improved metabolic fitness and increased stress resistance in a wide range of organisms. DR is often referred to as calorie restriction, implying that reduced energy intake is responsible for its widespread and evolutionarily conserved benefits. However, recent data indicate dietary amino acid restriction as a key mediator of DR benefits. In fruitflies, an imbalance in essential amino acid intake is thought to underlie longevity benefits of DR. In mammals, reduced dietary protein or essential amino acid intake can extend longevity, improve metabolic fitness and increase stress resistance. In the present paper we review two evolutionarily conserved signal transduction pathways responsible for sensing amino acid levels. The eIF2 α (eukaryotic initiation factor 2 α) kinase GCN2 (general amino acid control non-derepressible 2) senses the absence of one or more amino acids by virtue of direct binding to uncharged cognate tRNAs. The presence of certain amino acids, such as leucine, permits activation of the master growth regulating kinase TOR (target of rapamycin). These two signal transduction pathways react to amino acid deprivation by inhibiting general protein translation while at the same time increasing translation of specific mRNAs involved in restoring homeostasis. Together, these pathways may contribute to the regulation of longevity, metabolic fitness and stress resistance.